

Physics 7501 Useful Tips

Sharon Ye

1 Trigonometric Identities

Instead of memorizing all the trigonometric identities, you can derive them from

$$e^{i\theta} = \cos \theta + i \sin \theta \quad \text{and} \quad e^{i\alpha} e^{i\beta} = e^{i(\alpha+\beta)}$$

From these two expressions we get that

$$e^{ix} = \cos x + i \sin x \quad \text{and} \quad e^{-ix} = \cos x - i \sin x.$$

$\Rightarrow$

$$\boxed{\cos x = \frac{e^{ix} + e^{-ix}}{2}} \quad \text{and} \quad \boxed{\sin x = \frac{e^{ix} - e^{-ix}}{2i}}$$

1.1 Addition and Subtraction Formulas

$$e^{i(a+b)} = e^{ia} e^{ib}.$$

Using Euler's formula,

$$\begin{aligned} \cos(a+b) + i \sin(a+b) &= (\cos a + i \sin a)(\cos b + i \sin b) \\ &= \cos a \cos b - \sin a \sin b + i(\sin a \cos b + \cos a \sin b) \end{aligned}$$

Now equate real and imaginary parts:

$$\boxed{\cos(a+b) = \cos a \cos b - \sin a \sin b}$$

$$\boxed{\sin(a+b) = \sin a \cos b + \cos a \sin b}.$$

To get subtraction formulas, simply replace $b \rightarrow -b$:

$$\boxed{\cos(a-b) = \cos a \cos b + \sin a \sin b}$$

$$\boxed{\sin(a-b) = \sin a \cos b - \cos a \sin b}.$$

1.2 Double/Half Angle Formulas

$$e^{2ix} = (e^{ix})^2.$$

$\Rightarrow$

$$\begin{aligned} \cos 2x + i \sin 2x &= (\cos x + i \sin x)^2 \\ &= \cos^2 x - \sin^2 x + 2i \sin x \cos x \end{aligned}$$

$\Rightarrow$

$$\boxed{\cos 2x = \cos^2 x - \sin^2 x} \quad , \quad \boxed{\sin 2x = 2 \sin x \cos x}$$

Using the Pythagorean identity

$$1 = e^{ix} e^{-ix} = (\cos x + i \sin x)(\cos x - i \sin x) = \cos^2 x + \sin^2 x.$$

and combining it with

$$\cos 2x = \cos^2 x - \sin^2 x$$

we get the other forms of the double-angle identity:

$$\boxed{\cos 2x = 2 \cos^2 x - 1} \quad , \quad \boxed{\cos 2x = 1 - 2 \sin^2 x}$$

Solving these gives the half-angle/power-reduction formulas:

$$\boxed{\cos^2 x = \frac{1 + \cos 2x}{2}}, \quad \boxed{\sin^2 x = \frac{1 - \cos 2x}{2}}.$$

1.3 Product to Sum Identities

Use the exponential representation e.g.

$$\begin{aligned} \cos a \cos b &= \frac{e^{ia} + e^{-ia}}{2} \frac{e^{ib} + e^{-ib}}{2} \\ &= \frac{1}{4} \left(e^{i(a+b)} + e^{i(a-b)} + e^{-i(a-b)} + e^{-i(a+b)} \right) \\ &= \frac{1}{4} [2 \cos(a+b) + 2 \cos(a-b)] \end{aligned}$$

$\Rightarrow$

$$\boxed{\cos a \cos b = \frac{1}{2} [\cos(a+b) + \cos(a-b)]}$$

The same procedure gives

$$\boxed{\sin a \sin b = \frac{1}{2} [\cos(a-b) - \cos(a+b)]} \quad , \quad \boxed{\sin a \cos b = \frac{1}{2} [\sin(a+b) + \sin(a-b)]}.$$

1.4 Multiple Angle Identities and de Moivre's Formula

$$(e^{ix})^n = e^{inx} \Rightarrow \boxed{(\cos x + i \sin x)^n = \cos(nx) + i \sin(nx)}.$$

This is de Moivre's formula. For example, for $n = 3$,

$$(\cos x + i \sin x)^3 = \cos 3x + i \sin 3x.$$

Expand the left side. Its real part is

$$\cos^3 x - 3 \cos x \sin^2 x.$$

Thus

$$\cos 3x = \cos^3 x - 3 \cos x \sin^2 x.$$

Using $\sin^2 x = 1 - \cos^2 x$,

$$\cos 3x = 4 \cos^3 x - 3 \cos x.$$

The imaginary part similarly gives

$$\sin 3x = 3 \sin x - 4 \sin^3 x.$$

So all the n -angle identities come from one equation.

2 Bloch Sphere

The Bloch sphere is useful because it turns a two-level quantum state into geometry: measurement probabilities become projections, unitary evolution becomes rotation, and mixedness becomes distance from the center.

2.1 Visualizing a state as a point on the sphere

Choose $|0\rangle = |\uparrow_z\rangle$ and $|1\rangle = |\downarrow_z\rangle$. Any normalized pure qubit state, after removing an overall phase, can be written as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$$

Its Bloch vector is

$$\mathbf{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

which is the unit radial vector in spherical coordinates. Here θ is the polar angle measured from $+z$, and ϕ is the azimuthal angle.

NOTE The half-angle in the ket matters! Some useful landmarks:

- (i) $|0\rangle$ and $|1\rangle$ are the north and south poles.
- (ii) $(|0\rangle \pm |1\rangle)/\sqrt{2}$ point along $\pm x$.
- (iii) $(|0\rangle \pm i|1\rangle)/\sqrt{2}$ point along $\pm y$.

Orthogonal kets correspond to opposite points, not perpendicular Bloch vectors. Multiplying a ket by an overall phase leaves its point unchanged; changing the relative phase generally moves it.